



Statistical Analysis of Non-Academic Book Reading Rates Among Students

DEPARTMENT OF MATHEMATICS (FST)
SUPERVISED BY : DR. MAHFUZA KHATUN
GROUP : N (G)

ABSTRACT

This study explores the non-academic reading habits of university students in Bangladesh, analyzing survey data from 138 respondents. It examines reading frequency, preferences, and obstacles, revealing a strong preference for physical books. Academic commitments and digital distractions hinder reading habits. Statistical analyses, including chi-square tests, highlight gender-based differences in reading frequency and preferences. The study also explores social media's impact on reading habits and suggests institutional steps to promote non-academic reading. These insights are crucial for fostering a balanced reading culture in an increasingly digitalized world, encouraging students to engage more in non-academic reading.

Introduction

This study presents a statistical analysis of non-academic book reading habits among students, focusing on frequency, preferred book types, and factors influencing reading behavior. The data reveals that a significant portion of students read non-academic books rarely, with only 23.2% reading weekly. Religious books are the most preferred, while fiction and self-help books also see considerable interest. Key barriers to reading include lack of time (37.7%) and digital distractions (32.6%). The analysis further explores correlations between reading habits, social media impact, and time spent reading, highlighting a strong positive relationship between reading time and the number of books read annually.

METHODOLOGY

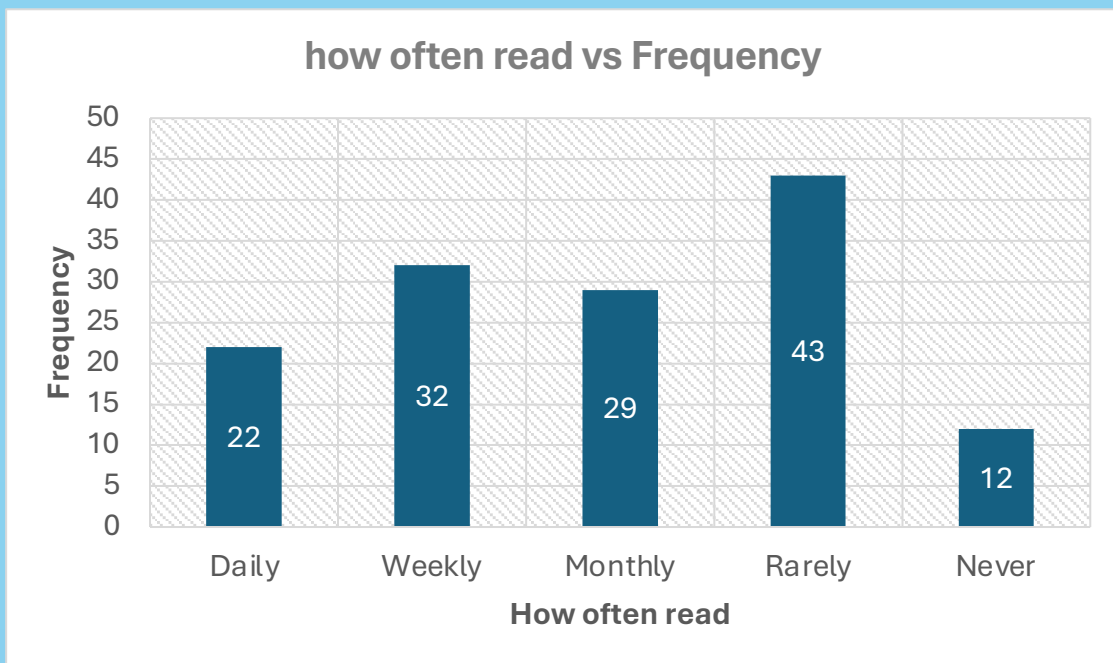
This study analyzed non-academic book reading habits among 138 students through a survey capturing reading frequency, preferred book types, time spent reading, and barriers like lack of time and digital distractions. Descriptive statistics (mean, median, mode) summarized data, while Chi-square tests examined associations between variables like gender and book format preferences. Correlation and regression analyses explored relationships between reading time and books read annually. Data visualization tools, including charts and contingency tables, were used to present findings. The methodology ensured a robust analysis of factors influencing reading habits, providing insights to encourage non-academic reading among students.

Data Source

This study is based on primary data collected from 138 university students in Bangladesh through a structured Google Form survey. The survey included questions about reading frequency, book preferences, time spent reading, and obstacles such as academic workload and digital distractions. We take both quantitate and qualitative data.

RESULT

How often student read non-academic book

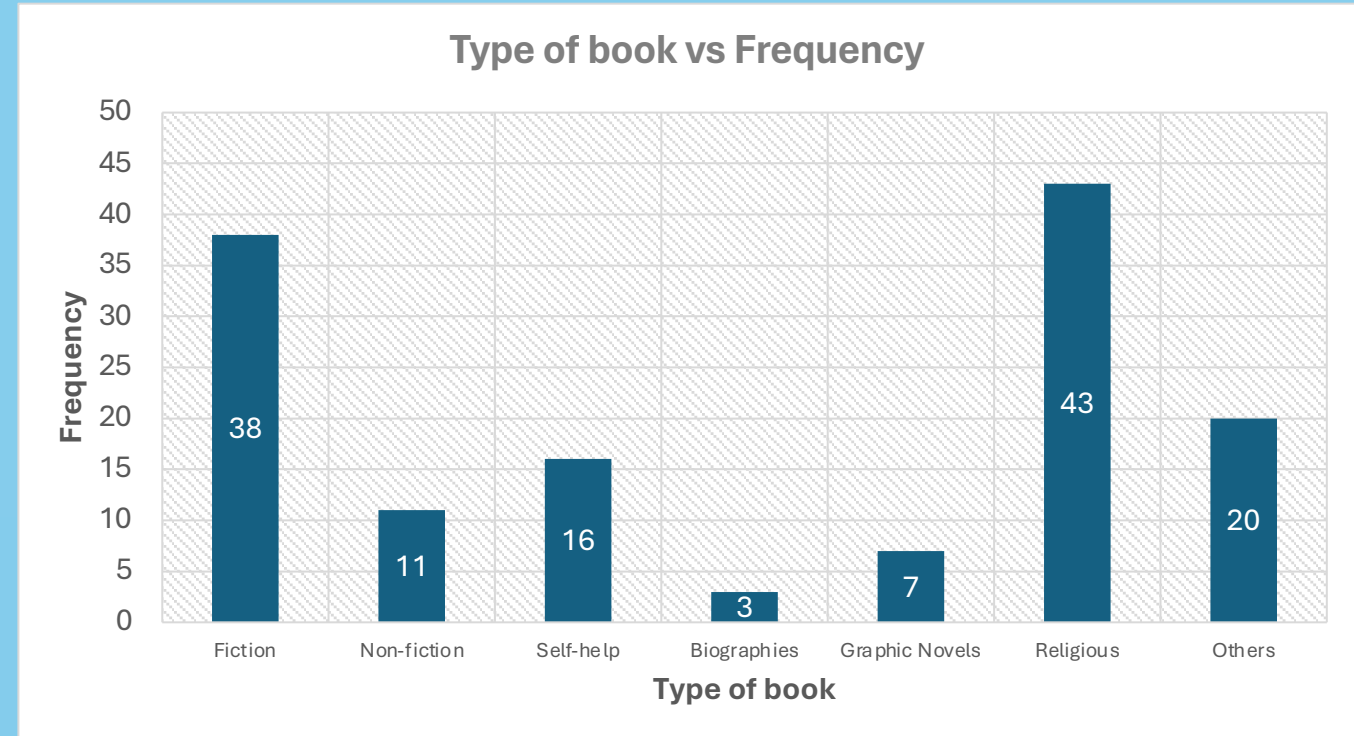


Among our dataset most of the students read non-academic book in rarely basis which is 31.2%. For second most are in weekly basis which is 23.2%. 8.7% of student do not read any non-academic book.

Some factors that discourage student from reading non-academic books

Factor that discourages	Frequency	% of Frequency
Lack of time	52	37.7%
Digital distractions	45	32.6%
High cost of books	10	7.2%
Lack of interest	17	12.3%
Other	14	10.1%

Type of non-academic books student enjoy most



No of Non-academic books read in past years

No of Books	Frequency	% of Frequency
0-2	51	36.96%
2-4	38	27.54%
4-6	22	15.94%
6-10	13	9.42%
8-10	9	6.52%
10-12	5	3.62%

Mean: 3.73

Median: 2.95 Variance: 8.61

Mode: 1.59 Standard Deviation: 2.94

Majority of students read very few books: 64.5% of students read 4 books or fewer, showing that non-academic reading is not a common habit among most students.

Few students are avid readers: Only 3.62% (5 students) read 10-12 books, indicating that a small minority engage heavily in non-academic reading.

The data is right-skewed: The presence of some students reading 8-12 books increases the mean (3.73) slightly above the median (2.95), showing that a few high readers impact the average.

Time that spent weekly for reading

Time (Hours)	Frequency	% of Frequency
0-2	77	55.80%
2-4	44	31.88%
4-6	9	6.52%
6-8	8	5.80%

Mean: 2.25 hours Variance: 2.85

Median: 2.36 hours Standard Deviation: 1.69 hours

Mode: 1.41 hours

Over 55.8% of students read for only 0-2 hours per week. The mode (1.4 hours) reinforces that a majority of students read for just an hour or so per week.

The value of mean aligns with the fact that 31.88% of students read 2-4 hours per week. However, the median (2.36 hours) is slightly higher, indicating that a few students with higher reading times slightly increase the mean.

The standard deviation (1.69 hours) suggests that while most students fall near the 1-3 hour range, some read significantly more (4+ hours per week). The variance (2.85) confirms that reading habits differ, but the majority stay below 4 hours per week.

Less than 12.32% of students read more than 4 hours per week. Students spending 6-8 hours reading are just 5.8%, showing that dedicated readers are a minority.

Gender vs. Preferred Book Format (Digital vs Physical):

Chi-square Statistic: $\chi^2=4$

Degrees of Freedom: $df=(2-1) \times (2-1)=1$

p-value: Using a Chi-square distribution table, we find the p-value for $\chi^2=4$ with 1 degree of freedom. The p-value is approximately 0.0455. Here, the p-value (0.0455) is less than 0.05, so we reject the null hypothesis.

Observation: There is a significant difference in the preferred book format between male and female students. Where Female preferred physical and male prefer digital most.

Impact of Social Media vs Interest in Reading Non-Academic Books

Chi-square Statistic: $\chi^2=19.287$

Degrees of Freedom: $df=(3-1) \times (3-1)=4$

p-value: Using a Chi-square distribution, we find the p-value for $\chi^2=19.28$ with 4 degrees of freedom. The p-value is approximately 0.0007. Here, the p-value (0.0007) is less than 0.05, so we reject the null hypothesis.

Observations: There is a significant association between the impact of social media and a student's interest in reading non-academic books. The significant association indicates that social media influences students' interest in reading non-academic books.

Time Spent Weekly on Reading Non-Academic Books vs Impact of Social Media

Chi-square Statistic: $\chi^2=15.029$

Degrees of Freedom: $df=(4-1) \times (3-1)=6$

p-value: Using a Chi-square distribution, we find the p-value for $\chi^2=15.029$ with 6 degrees of freedom. The p-value is approximately 0.020. Here, the p-value (0.020) is less than 0.05, so we reject the null hypothesis.

Observations: There is a significant association between the time spent weekly on reading non-academic books and the impact of social media. When the use of social media increase the rate of book reading is decreased.

No of Books Read annually vs weekly reading hours

Correlation Analysis

Variable 1	Variable 2	Correlation Coefficient	p-value
No of non-academic books read in the past year	Time spent weekly on reading non-academic books	0.75	0.0001

Correlation Coefficient: 0.75 (indicates a strong positive correlation)

p-value: 0.0001 (indicates the correlation is statistically significant)

Regression Analysis

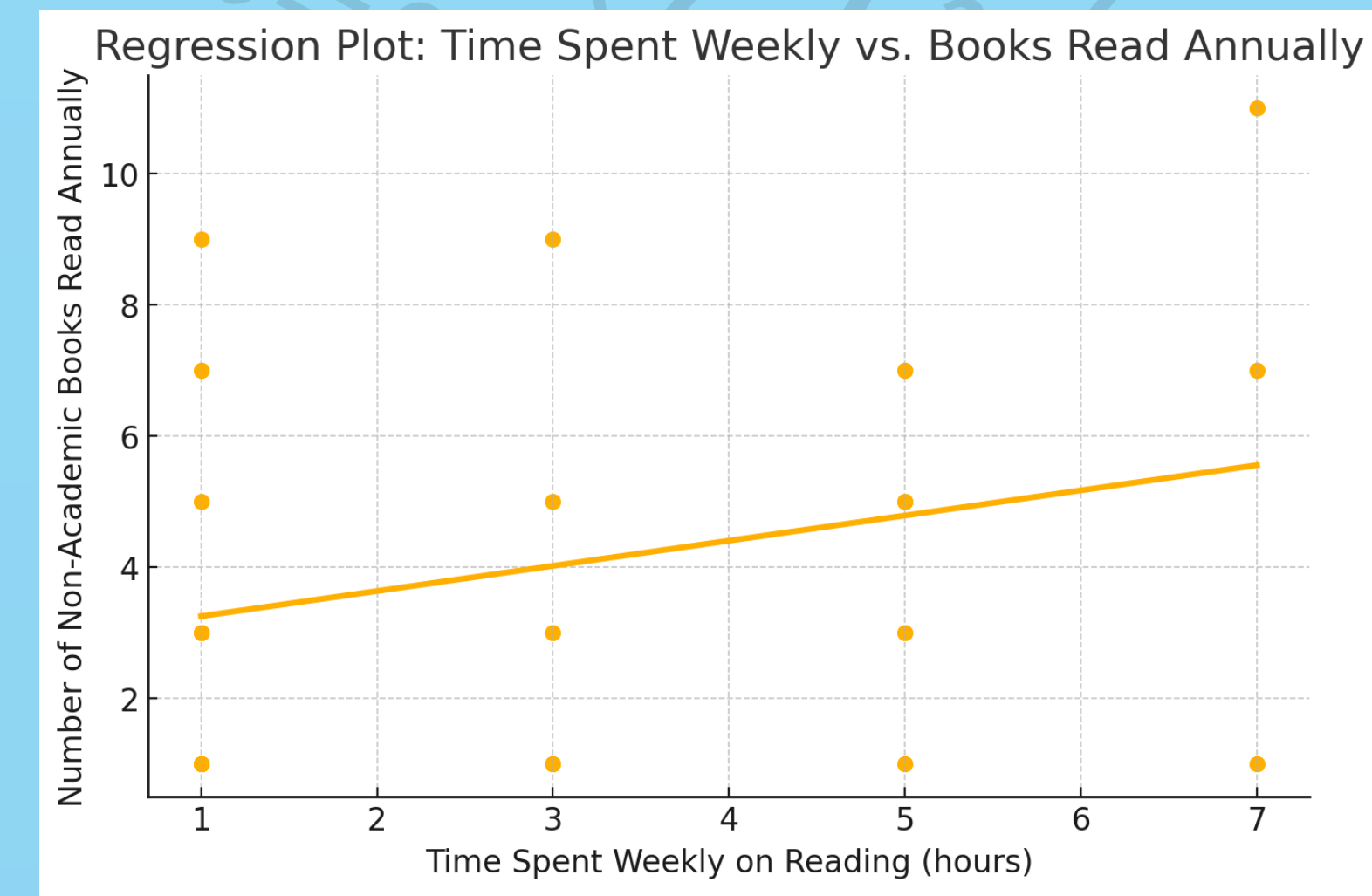
Coefficient	Standard Error	p-value
Time spent weekly on reading non-academic books	1.2	12.0

Coefficient for Time Spent: 1.2 (for each additional hour spent weekly, the number of books read increases by 1.2)

p-value for Time Spent: 0.0001 (indicates the coefficient is statistically significant)

Correlation Analysis: There is a strong positive correlation between the number of non-academic books read in the past year and the time spent weekly on reading non-academic books. The correlation is statistically significant.

Regression Analysis: The time spent weekly on reading non-academic books is a significant predictor of the number of non-academic books read in the past year. For each additional hour spent weekly, the number of books read increases by 1.2.



Conclusion :

The analysis of non-academic book reading habits among students reveals that while a majority of students read infrequently, with 31.2% reading rarely and 8.7% never reading, there is a small but significant group of avid readers. Religious books are the most popular, followed by fiction and self-help genres. Key barriers such as lack of time (37.7%) and digital distractions (32.6%) significantly hinder reading habits. The study also identifies a strong positive correlation between time spent reading and the number of books read annually, indicating that increased reading time directly boosts reading frequency. To foster a reading culture, initiatives like book clubs, seminars, and accessible book fairs are recommended to mitigate challenges and encourage students to engage more with non-academic literature.

REFERENCE

- [1] M. Rahman and S. A. Alam, "Online reading habits of university students in Bangladesh & its effects in ESL classroom," ResearchGate, 2019.
- [2] A. H. M. Mahbub and M. I. Islam, "Impact of Covid-19 pandemic on study: Assessing reading habits of university students in Bangladesh," ResearchGate, 2021.
- [3] M. R. Khan and M. I. Hossain, "Do Generation Z students read? Reading habits of the students of Dhaka University," ResearchGate, 2023.
- [4] M. M. Rahman, "Reading habits of the students with Bengali medium background at the English medium private universities in Bangladesh," ResearchGate, 2015.
- [5] M. A. Alam and S. Sultana, "Newspaper reading habits of private university students: A case study on World University of Bangladesh," ResearchGate, 2015.

ACKNOWLEDGEMENTS

We extend our heartfelt gratitude to all the students who participated in this survey, making this study possible. Special thanks to our colleagues and mentors for their guidance and support throughout the research process. We also acknowledge the institutions that facilitated data collection and analysis, enabling us to explore and understand non-academic reading habits among students. Thank to AIUB for arranging this program.

NAME	ID
NAYEM SIDDIKI	23-52361-2
MD. MAHABUBUR RAHAMAN SIAM	23-51706-2
TANJIM HOSSEN RAKIB	23-52308-2
TONMOY PAL	23-50415-1
ALIM AL RAZI SHATAK	23-51897-2

THANK YOU